Computer Networks

BRANCH: COMPUTER SCIENCE (CSE)
COURSE CODE: CS304

Subject Overview & Scope

This comprehensive document serves as the official compiled study notes for Computer Networks (CS304) under the Computer Science (CSE) department. It is structured into four core university syllabus units covering primary concepts, definitions, formulas, and advanced technical applications for university examinations and placements.

University Course Syllabus & Core Units

Unit 1: Introduction & Physical Layer

Data Communication, Network topologies, OSI vs TCP/IP Reference Model. Physical Layer: Transmission media (guided, unguided), line coding, multiplexing (FDM, TDM, WDM), switching.

Unit 2: Data Link Layer

Framing, Error control (Parity, CRC, Hamming code). Flow control (Stop-and-Wait, Sliding Window, Go-Back-N, Selective Repeat). MAC Sublayer: ALOHA, CSMA/CD, CSMA/CA, Ethernet, Token Ring.

Unit 3: Network Layer

Routing algorithms: Distance Vector (RIP), Link State (OSPF). Subnetting and Classless Addressing (CIDR). IPv4 vs IPv6. ARP, DHCP, ICMP protocols.

Unit 4: Transport & Application Layer

Connectionless vs Connection-oriented: UDP vs TCP. Flow control, congestion control in TCP. Application Layer: DNS, HTTP, HTTPS, FTP, SMTP, POP3, DHCP.

Key Formulas & Exam Pointers

CRITICAL FORMULA / PRINCIPLE:

*Nyquist Bit Rate = 2 * B * log2(L). Shannon Capacity = B * log2(1 + SNR). IPv4 Address Space = 2^32.*